Experimenting to Determine the Best Chunk Size and Overlap for the Retrieval of Workers' Statutory Benefit Information

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***Abstract*—This study explored how chunk size and overlap affect the retrieval performance of a FAISS-based Retrieval-Augmented Generation system using the *Workers' Statutory Monetary Benefits Handbook 2024 Edition*. Twelve chunking configurations were tested, combining chunk sizes of 300, 500, 700, and 1,000 characters with zero, 10%, and 20% overlap. Each configuration was evaluated using the same nine worker-oriented questions, evenly divided into easy, average, and difficult levels. Precision@3, Recall@3, and Mean Reciprocal Rank at 3 were used to measure retrieval quality, while the number of generated chunks and indexing time provided a practical view of efficiency. One noticeable pattern was that smaller chunks could isolate specific details, but they also tended to separate related conditions and computations. Larger chunks preserved more of the surrounding explanation, which became especially valuable for questions requiring evidence from several passages. Configuration C12, using a chunk size of 1,000 characters and an overlap of 200 characters, achieved the highest Precision@3 of 0.556 and Recall@3 of 0.944. Its MRR@3 of 0.685 also remained close to the highest recorded score of 0.704. Although C10 required fewer chunks, C12 offered more complete and consistent evidence across average and difficult questions. These findings suggest that the additional retrieval context provided by C12 was a reasonable trade-off for improved evidence coverage, making it the most suitable configuration for the selected handbook.**

***Keywords*—Retrieval-Augmented Generation, FAISS, chunk size, chunk overlap, information retrieval, Precision@3, Recall@3, MRR@3, Qwen3-1.7B, statutory monetary benefits.**

I. Introduction

Workers often need clear and accurate information about wages, holiday pay, premium pay, leave benefits, and other statutory compensation. Although these rules are available in official handbooks, finding the correct provision can still be challenging because the answer may depend on several conditions, exceptions, or related passages. Retrieval-Augmented Generation can make this information more accessible by dividing a document into chunks, converting them into embeddings, and retrieving the passages most relevant to a worker's question. During this process, chunk size and overlap become important design choices. Smaller chunks may provide focused information but can separate a benefit rate from its required conditions, while larger chunks may preserve context but include unrelated details. Overlap can help retain information near chunk boundaries, although it also increases redundancy and the number of chunks that must be processed.

This study evaluated twelve chunk-size and overlap configurations using the *Handbook on Workers' Statutory Monetary Benefits, 2024 Edition* and nine worker-oriented questions divided into easy, average, and difficult levels. Precision@3, Recall@3, MRR@3, total chunks, and indexing time were examined to identify the configuration that provided the most dependable balance of evidence completeness, retrieval consistency, and efficiency.

II. Model Background

The language model initially proposed for the activity was GPT-OSS-20B. However, its GPU requirements could not be accommodated by the resources available in the free tier of Google Colab. This limitation encouraged the selection of a smaller model that could still support the main purpose of the experiment without requiring access to higher-memory hardware. Qwen3-1.7B was therefore used as the response-generation model and executed using a T4 GPU. Its smaller size made it more practical for the available environment, while its instruction-following capability allowed it to answer questions using the passages supplied by the retrieval process. The model was instructed to rely only on the retrieved context and to avoid adding unsupported information, which was especially important because the source document contains specific rates, conditions, and exceptions affecting workers' benefits.

Document passages and user questions were converted into numerical representations using the paraphrase-multilingual-MiniLM-L12-v2 embedding model. FAISS then compared these embeddings to retrieve the three most relevant chunks, while a reranking stage helped place stronger evidence closer to the top. In this arrangement, Qwen3-1.7B did not search the handbook directly. Instead, it explained the evidence selected by the retrieval pipeline, making the accuracy of the final response closely connected to the quality and completeness of the retrieved chunks.

III. Experimental Methodology

A. Dataset Description

The dataset used in this study was the *Handbook on Workers' Statutory Monetary Benefits, 2024 Edition*, published by the Department of Labor and Employment through the Bureau of Working Conditions. The handbook contains 81 PDF pages covering wages, holiday and premium pay, leave benefits, thirteenth-month pay, retirement benefits, and other statutory programs for workers in the Philippines. It contains direct numerical rules as well as conditions, exceptions, legal references, and sample computations. Nine worker-oriented questions were prepared from the document, consisting of three easy, three average, and three difficult questions. This distribution made it possible to observe retrieval performance for direct facts, related conditions, and more complex questions requiring evidence from several passages.

B. Chunking Configurations

Twelve chunking configurations were tested using chunk sizes of 300, 500, 700, and 1,000 characters. Each size was paired with overlap values of 0%, 10%, and 20%. For example, the 1,000-character configurations used overlaps of 0, 100, and 200 characters. A new FAISS index was created for every configuration to ensure that chunks from one setting were not reused in another. The same source document, embedding model, test questions, top-three retrieval setting, reranking process, language model, and prompt were maintained throughout the experiment. Keeping these elements consistent helped isolate the effects of chunk size and overlap. The number of generated chunks and indexing time were also recorded because a configuration should remain practical in addition to producing accurate retrieval results.

C. Evaluation Method

Retrieval performance was evaluated using Precision@3, Recall@3, and Mean Reciprocal Rank at 3. Precision@3 measured the proportion of the three retrieved chunks that contained useful evidence, while Recall@3 measured how much of the required evidence was recovered. MRR@3 considered how early the first relevant chunk appeared in the ranked results. Expected answers and supporting passages were identified before evaluating the configurations. Questions requiring several facts were divided into evidence units so that complete and partial retrieval could be distinguished. A retrieved chunk was considered relevant only when it genuinely supported the answer, rather than simply mentioning similar words or topics. The same relevance criteria were applied to all twelve configurations, allowing their overall and difficulty-level results to be compared consistently.

IV. Results

The twelve configurations produced noticeable differences in both retrieval quality and the number of generated chunks. As the chunk size increased, the total number of chunks generally decreased. The strongest overall results came from the larger chunk configurations, particularly C12, which used a chunk size of 1,000 characters and an overlap of 200 characters. It achieved the highest Precision@3 of 0.556 and the highest Recall@3 of 0.944. C8 and C10 recorded the highest MRR@3 of 0.704, showing that their first relevant chunks often appeared slightly earlier. Still, C12 remained close with an MRR@3 of 0.685 while recovering a larger portion of the required evidence.

TABLE I. Overall Retrieval Performance

Config.	Chunk Size	Overlap	Total Chunks	Precision@3	Recall@3	MRR@3
C1	300	0	517	0.259	0.444	0.37
C2	300	30	569	0.296	0.463	0.5
C3	300	60	633	0.222	0.426	0.5
C4	500	0	323	0.37	0.519	0.37
C5	500	50	359	0.407	0.519	0.444
C6	500	100	398	0.37	0.519	0.537
C7	700	0	242	0.481	0.889	0.63
C8	700	70	269	0.444	0.778	0.704
C9	700	140	295	0.444	0.685	0.648
C10	1,000	0	180	0.444	0.778	0.704
C11	1,000	100	199	0.519	0.778	0.648
C12	1,000	200	217	0.556	0.944	0.685

Looking at the questions by difficulty provided a clearer view of how the configurations behaved. Several configurations performed well on easy questions because the required facts were usually available in one passage. The differences became more visible for average and difficult questions, where conditions and supporting evidence had to remain together. C12 maintained complete recall for easy and average questions and reached a difficult-question recall of 0.833. This pattern suggested that its performance was not dependent on only one difficulty level.

TABLE II. Performance On Easy Questions

Config.	Precision@3	Recall@3	MRR@3
C1	0.556	1	0.667
C2	0.556	1	0.833
C3	0.444	1	1
C4	0.444	0.667	0.5

Config.	Precision@3	Recall@3	MRR@3
C5	0.556	0.667	0.5
C6	0.556	0.667	0.667
C7	0.556	1	0.778
C8	0.444	0.667	0.667
C9	0.444	0.667	0.667
C10	0.556	1	0.667
C11	0.667	1	0.611
C12	0.556	1	0.611

TABLE III. PERFORMANCE ON AVERAGE QUESTIONS

Config.	Precision@3	Recall@3	MRR@3
C1	0	0	0
C2	0.111	0.167	0.333
C3	0.111	0.167	0.333
C4	0.333	0.5	0.333
C5	0.333	0.5	0.5
C6	0.222	0.5	0.5
C7	0.444	1	0.667
C8	0.333	0.833	0.833
C9	0.333	0.667	0.5
C10	0.333	0.667	0.667
C11	0.333	0.667	0.667
C12	0.556	1	0.667

TABLE IV. PERFORMANCE ON DIFFICULT QUESTIONS

Config.	Precision@3	Recall@3	MRR@3
C1	0.222	0.333	0.444
C2	0.222	0.222	0.333
C3	0.111	0.111	0.167
C4	0.333	0.389	0.278
C5	0.333	0.389	0.333
C6	0.333	0.389	0.444
C7	0.444	0.667	0.444
C8	0.556	0.833	0.611
C9	0.556	0.722	0.778
C10	0.444	0.667	0.778
C11	0.556	0.667	0.667
C12	0.556	0.833	0.778

The results show that no configuration dominated every individual score. However, C12 provided the strongest overall balance. It led both overall precision and recall, matched the best difficult-question results, and remained competitive in ranking relevant evidence. C10 was more efficient because it generated only 180 chunks, compared with 217 for C12, but its lower recall indicated that some required evidence was more likely to be missed.

V. DISCUSSION

The experiment showed that chunk size influenced retrieval more consistently than overlap. The 300-character configurations occasionally isolated specific details, but they also separated percentages, eligibility conditions, and related computations. This fragmentation was especially noticeable in average and difficult questions, where several details had to remain connected. In contrast, the 700- and 1,000-character chunks generally preserved enough surrounding context to support a complete answer. C12 demonstrated this advantage by retrieving complementary evidence for questions involving several benefit rules. However, its Precision@3 of 0.556 also showed that larger chunks could still contain related but unnecessary information.

Increasing overlap did not automatically improve retrieval. It always increased the number of generated chunks, but its effect on quality depended on the chunk size. The 200-character overlap improved the 1,000-character configuration, allowing C12 to reach a Recall@3 of 0.944. The same pattern was not observed consistently among the smaller configurations. Another interesting comparison involved C8 and C10, which achieved identical overall retrieval scores even though C10 produced 89 fewer chunks. C10 was therefore more efficient, but C12 was more dependable when evidence completeness was considered. For a handbook containing conditions and exceptions that can affect the meaning of a benefit rule, retrieving complete evidence was considered more important than the relatively small efficiency advantage.

The experiment was limited to one handbook, nine questions, a top-three retrieval setting, and one embedding and reranking approach. Relevance judgments were based on manually identified evidence units, which may still involve some interpretation. Indexing time was also measured from only one run per configuration, so the recorded differences should not be treated as a complete performance benchmark. Future testing could include more worker questions, repeated timing measurements, other document types, and different retrieval depths. Even with these limitations, the results offered a useful observation: a chunking configuration should not be selected from one metric alone, because the strongest setting must balance ranking quality, evidence coverage, consistency, and practical efficiency.

VI. Final Recommendation

Based on the experimental results, C12 is recommended as the final chunking configuration for the *Handbook on Workers' Statutory Monetary Benefits, 2024 Edition*. This configuration uses a chunk size of 1,000 characters and an overlap of 200 characters. It achieved the highest overall Precision@3 of 0.556 and Recall@3 of 0.944, while its MRR@3 of 0.685 remained close to the highest observed score of 0.704. It also performed consistently across the three difficulty levels, reaching complete recall for easy and average questions and a Recall@3 of 0.833 for difficult questions. These results suggest that C12 was better able to preserve the conditions, rates, and supporting passages needed for complete answers.

C10 remained a practical alternative because it generated only 180 chunks compared with the 217 chunks produced by C12. It also achieved a slightly higher MRR@3, indicating that its first relevant result sometimes appeared earlier. However, its lower overall recall and weaker evidence coverage for average and difficult questions presented a greater possibility of missing an important condition or supporting passage. Since accuracy and evidence completeness are particularly important when answering questions about workers' statutory benefits, the additional 37 chunks produced by C12 were considered a reasonable trade-off. For this dataset, C12 provided the most suitable balance of retrieval quality, consistency, and efficiency.

VII. Conclusion

This study examined how chunk size and overlap influenced the retrieval performance of a FAISS-based RAG system using the *Handbook on Workers' Statutory Monetary Benefits, 2024 Edition*. The results showed that smaller chunks could retrieve focused details, but they often separated information that needed to remain connected, such as benefit rates, eligibility conditions, and exceptions. Larger chunks generally preserved this context more effectively. Overlap was also useful in certain settings, although increasing it did not consistently improve every retrieval metric and always produced additional chunks.

Among the twelve configurations, C12, with a chunk size of 1,000 characters and an overlap of 200 characters, provided the strongest overall balance. It achieved the highest Precision@3 and Recall@3 while remaining competitive in MRR@3 and performing consistently across the three difficulty levels. What became clear from the experiment was that the best configuration could not be selected from a single score alone. Evidence completeness, question difficulty, consistency, and computational efficiency all needed to be considered together. For this handbook, preserving the complete context of workers' benefit rules was worth the modest increase in the number of chunks. The findings provide a practical basis for using C12 as the final configuration and for continuing the development of a more accessible and dependable labor-benefit question-answering system.

References

[1] DOLE Bureau of Working Conditions, "Handbook on Workers' Statutory Monetary Benefits," 2024 Edition. Available: bwc.dole.gov.ph